ECE SKILLPATH

ECE Skill & Career Roadmap Platform

Mini Project Documentation

Developed using Base44 AI

Live Application: succinct-ece-path-pro.base44.app

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ABSTRACT

ECE SkillPath is a web-based skill development and career guidance platform designed for Electronics and Communication Engineering students. The application helps students understand their current technical skills, explore ECE career opportunities, identify skill gaps, follow personalized learning roadmaps, discover project ideas, and monitor their career readiness.

The platform was developed using Base44 AI, which supports natural-language app building, database management, authentication, responsive design and deployment. The project addresses a common problem among ECE students: uncertainty about which technical skills to learn and which career path to follow.

1. INTRODUCTION

Electronics and Communication Engineering provides opportunities in embedded systems, VLSI, PCB design, IoT, telecommunications, semiconductor technology, robotics, automotive electronics, software and government technical services. Students often find it difficult to connect academic subjects with industry roles and required practical skills.

ECE SkillPath provides a centralized platform that connects academic skills with career requirements and practical learning.

2. PROBLEM STATEMENT

Many ECE students have difficulty selecting a suitable career path, identifying the skills required for a target role, finding suitable projects and tracking their preparation. Information is often distributed across different websites, courses and documents. There is a need for a single platform that connects an ECE student's current skills with a practical career roadmap.

3. OBJECTIVES

- Develop a web platform specifically for ECE students.
- Assess and organize students' technical skills.
- Provide multiple ECE career pathways.
- Identify gaps between current and required skills.
- Generate structured learning roadmaps.
- Recommend suitable ECE projects and certifications.
- Help students track career-readiness progress.

4. SCOPE OF THE PROJECT

The platform can be used by B.Tech and diploma ECE students, fresh graduates and learners preparing for core electronics or related software careers. It can be extended with live job APIs, learning-resource integrations, advanced AI recommendations, institution dashboards and real-time analytics.

5. SYSTEM FEATURES

Student Profile: Stores degree, branch, year, CGPA, technical skills, tools, certifications, projects and career interests.

Skill Assessment: Organizes skills into Digital Electronics, Analog Electronics, Embedded Systems, Programming, PCB Design, VLSI, Communication, Control Systems, Signals and Systems, Network Theory and IoT.

Career Paths: Includes Embedded Systems, PCB Design, VLSI/RTL, Verification, IoT, Hardware Design, Semiconductor, RF/Telecom, Automotive Electronics, Robotics, Software and Government technical careers.

Skill Gap Analysis: Compares current skills with the expected skills for a selected target role.

Learning Roadmap: Provides staged learning plans from fundamentals to advanced topics and projects.

Project Recommendations: Provides ECE project ideas with difficulty, components, tools, skills and development stages.

Progress Tracker: Displays learning, project, certification and career-readiness progress.

6. TECHNOLOGY USED

Base44 AI: AI-powered application development and deployment.

Web Application: User interface and interaction.

Base44 Database: Application data management.

Authentication: User sign-up/login and user management.

Responsive UI: Desktop, tablet and mobile compatibility.

GitHub: Source-code repository and version control after export.

7. SYSTEM ARCHITECTURE

Architecture Flow: User → Web Interface → Authentication → Application Logic → Database → Dashboard / Recommendations / Roadmap / Progress.

The application uses a web interface connected to backend data and application logic. Student profiles, skills, career paths, roadmaps, projects and progress are maintained as structured application data.

8. MODULE DESCRIPTION

Authentication Module: Handles user access.

Profile Module: Maintains academic and career details.

Skill Module: Records technical skill levels.

Career Explorer: Displays ECE career options and requirements.

Skill Gap Module: Compares current and required skills.

Roadmap Module: Provides staged learning plans.

Project Module: Recommends practical ECE projects.

Progress Module: Tracks learning and career readiness.

9. WORKING PROCEDURE

1. Open the ECE SkillPath web application.
2. Create or access a student profile.
3. Enter academic details and existing technical skills.
4. Complete the skill assessment.
5. Select a preferred career path.
6. View the skill-gap analysis.
7. Follow the recommended learning roadmap.
8. Explore projects, certifications and exam preparation.
9. Mark completed learning activities.
10. Monitor progress through the dashboard.

10. DATABASE DESIGN

Student: ID, name, degree, branch, year, CGPA, interests.

Skill: ID, category, name, level.

Career Path: ID, title, description, required skills.

Roadmap: ID, career, stages, topics and status.

Project: ID, title, difficulty, components and tools.

Certification: ID, title, category and status.

Exam: ID, title, subjects and preparation status.

Progress: Student ID, skill progress, project progress and readiness.

11. TESTING

Application Loading: Verify that the live application opens correctly.

Profile Entry: Verify that profile information can be entered and saved.

Skill Selection: Verify that selected skills are recorded.

Career Selection: Verify that relevant career information appears.

Skill Gap: Verify that current and required skills are compared.

Roadmap: Verify that completed topics update progress.

Navigation: Verify that all major pages open correctly.

Responsive Design: Verify the interface on laptop, tablet and mobile screen sizes.

12. ADVANTAGES

- ECE-specific career guidance in one platform.
- Helps students identify skill gaps.
- Provides structured learning roadmaps.
- Connects theory with practical project development.
- Tracks student progress.
- Provides a professional and responsive interface.
- Can be expanded with additional AI and external integrations.

13. LIMITATIONS

- Career recommendations depend on the quality of available data.
- Live job and internship information requires suitable external integrations or manually maintained data.
- Skill assessment results are indicative and are not formal professional certification.
- Advanced AI features can depend on the capabilities and plan of the platform.

14. FUTURE ENHANCEMENTS

- Integrate verified live internship and job APIs.
- Add AI-based resume analysis.
- Add personalized interview preparation.
- Add ECE coding and circuit practice modules.
- Add mentor/student interaction.
- Add college-level placement analytics.
- Add multilingual support including Telugu.
- Improve AI-based recommendations using student performance data.

15. CONCLUSION

ECE SkillPath provides a practical solution for helping ECE students plan and monitor their technical career development. The application brings together skill assessment, career exploration, skill-gap analysis, learning roadmaps, projects and progress tracking.

Developing the project with Base44 AI demonstrates how natural-language application development can be used to create and deploy a functional web application efficiently. The platform can be extended into a larger career-development system with real-time data and advanced AI assistance.

16. PROJECT LINK

Live Application: https://succinct-ece-path-pro.base44.app

GitHub Repository: Add the repository URL after exporting the application code to GitHub.

17. REFERENCES

- Base44 AI platform and official product documentation.
- General ECE career and technical skill frameworks used to structure the application.